S5 Wave-pinning: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	8.617e-07	0.0002667	0.9999 / 0.9999	True / True	none
19	width_7.6	mesh401	True	8.617e-07	0.0002667	0.9999 / 0.9999	True / True	none
39	width_0.95	mesh401	True	0.0009561	0.002449	0.09274 / 0.09184	True / True	none
39	width_7.6	mesh401	True	1.56e-09	0.00112	1 / 1	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0001612	0.0002667	0.9999 / 0.9999	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.001581	0.00112	1 / 1	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.

S5 Goryachev: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	4.417e-12	0.003297	1 / 1	True / True	none
19	width_7.6	mesh401	True	1.311e-07	0.003296	1 / 1	True / True	none
39	width_0.95	mesh401	True	0	0.008964	1 / 1	True / True	none
39	width_7.6	mesh401	True	0	0.008964	1 / 1	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0006152	0.003297	-1 / -1	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.00109	0.008964	-1 / -1	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.

S5 deBelly: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	5.627e-05	0.0005137	0.9277 / 0.9278	True / True	none
19	width_0.95	quarter step201	True	9.114e-05	0.0003046	0.9277 / 0.9277	True / True	none
19	width_7.6	mesh401	True	5.375e-05	0.0004833	0.9278 / 0.9279	False / False	none
19	width_7.6	quarter step201	True	0.0001901	0.001004	0.9278 / 0.9277	False / True	tail
39	width_0.95	mesh401	True	3.826e-05	0.008586	0.9209 / 0.9209	True / True	none
39	width_0.95	quarter step201	False	0.0002683	0.02642	0.9209 / 0.9209	True / True	NC
39	width_7.6	mesh401	True	9.478e-05	0.008585	0.9209 / 0.9209	True / True	none
39	width_7.6	quarter step201	False	0.000661	0.02648	0.9209 / 0.9209	True / True	NC
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0008136	0.0005952	0.9276 / 0.9276	False / False	none
19	4 pulses, gap4, ON1.5τ	quarter step201	True	0.05893	0.001933	0.9276 / 0.9277	False / True	tail, eligibility, D
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.02738	0.0554	0.9165 / 0.9133	False / False	fields
39	4 pulses, gap4, ON1.5τ	quarter step201	False	0.7739	0.08341	0.9165 / 0.9209	False / True	NC, tail, fields, D, bias
19	width_0.95	qualified quarter step201	True	9.114e-05	0.0003046	0.9277 / 0.9277	True / True	none
19	width_7.6	qualified quarter step201	True	0.0001901	0.001004	0.9278 / 0.9277	False / True	tail
39	width_0.95	qualified quarter step201	True	0.0002655	0.02642	0.9209 / 0.9209	True / True	none
39	width_7.6	qualified quarter step201	True	0.000646	0.02648	0.9209 / 0.9209	True / True	none
19	4 pulses, gap4, ON1.5τ	qualified quarter step201	True	0.05893	0.001933	0.9276 / 0.9277	False / True	tail, eligibility, D
39	4 pulses, gap4, ON1.5τ	qualified quarter step201	True	0.775	0.08341	0.9165 / 0.9209	False / True	tail, fields, D, bias

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules. Both exact-native-start quarter-step diagnostics and separately qualified quarter-step basals are retained. The L39 mesh401 train has native-field endpoint sensitivity; qualified quarter-step trains preserve early bias loss and final fate but later-gap timing, eligibility and tail status differ.

S5 Holmes 3: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	1.153e-05	0.000299	0.9963 / 0.9964	True / True	none
19	width_7.6	mesh401	True	1.153e-05	0.000299	0.9963 / 0.9964	True / True	none
39	width_0.95	mesh401	True	0.001687	0.00499	-0.2697 / -0.2714	True / True	none
39	width_7.6	mesh401	True	0.0003869	0.001925	0.3536 / 0.3533	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0003293	0.000299	0.9963 / 0.9964	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules. The requested L39 train has no sampled commitment calibration through100,000 ON; its numerical companion is explicitly unavailable. It is not replaced with a forced-cap schedule.

S5 Spring: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	4.37e-06	1.62e-05	0.9964 / 0.9964	True / True	none
19	width_7.6	mesh401	True	4.37e-06	1.62e-05	0.9964 / 0.9964	True / True	none
39	width_0.95	mesh401	True	2.082e-06	6.576e-05	0.9997 / 0.9997	True / True	none
39	width_7.6	mesh401	True	2.082e-06	6.576e-05	0.9997 / 0.9997	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0006407	1.62e-05	0.9964 / 0.9964	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.02262	6.576e-05	0.9997 / 0.9997	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.